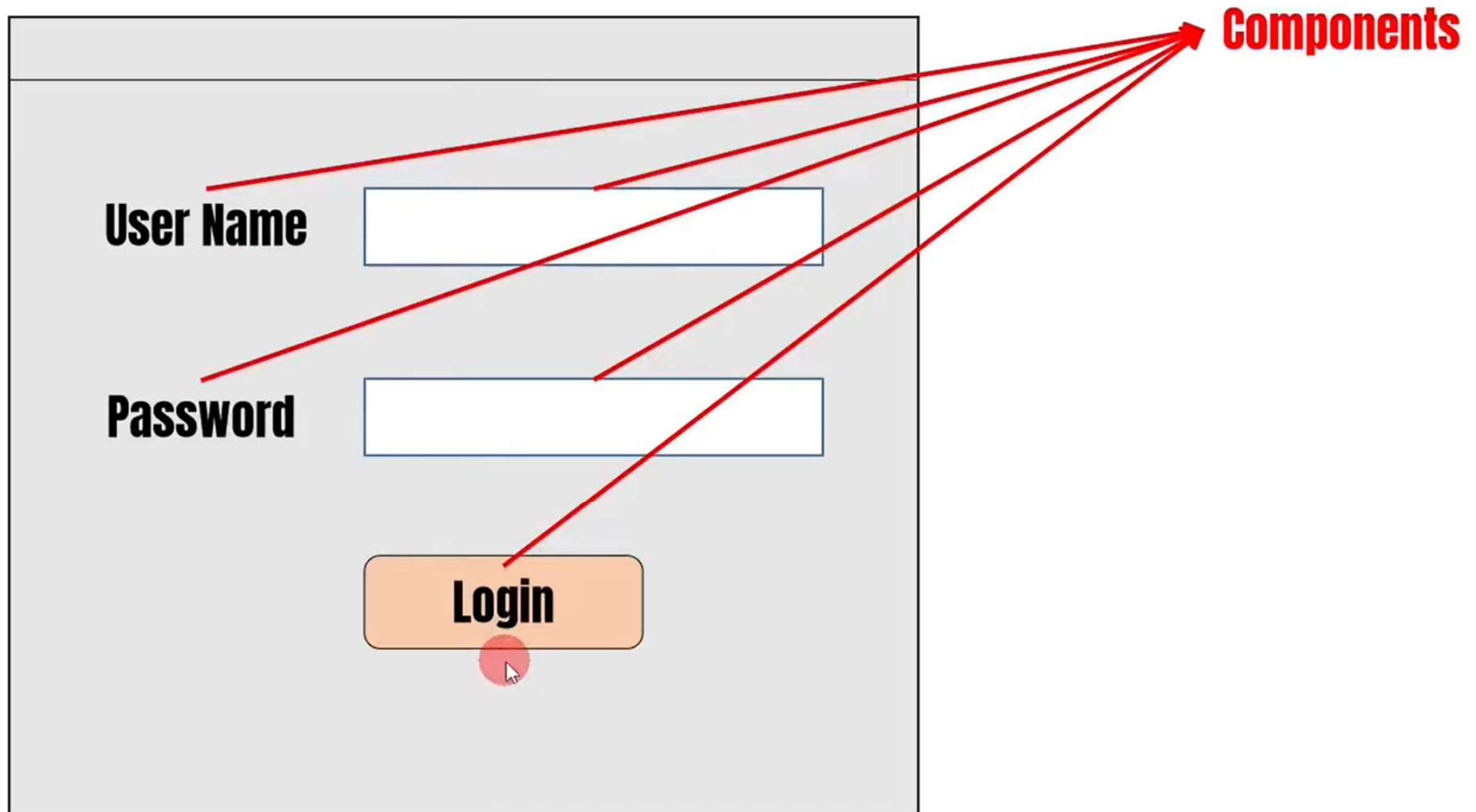


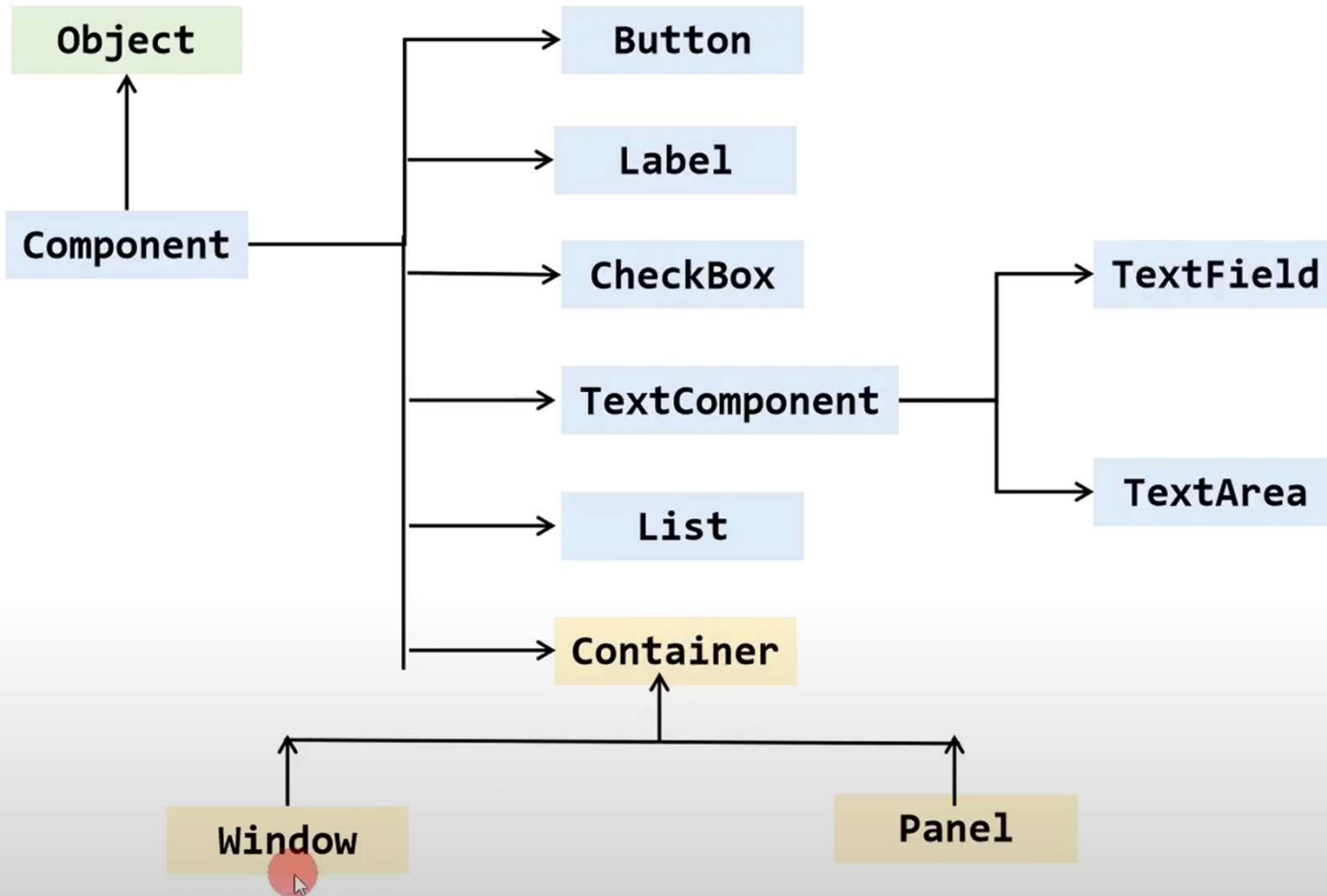
Introduction to AWT

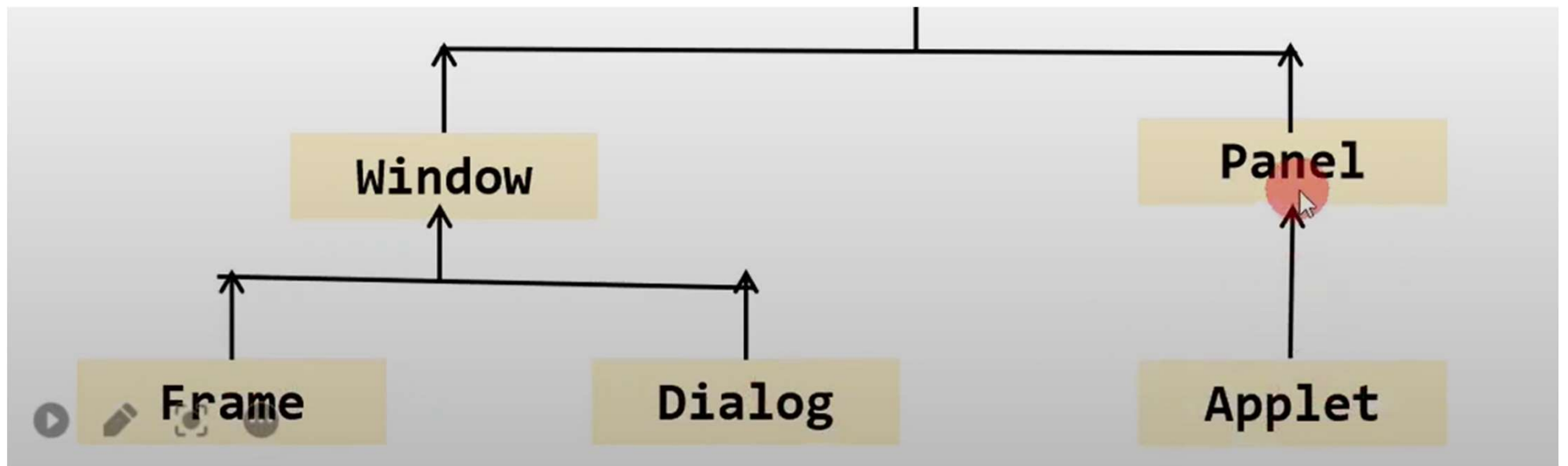
- AWT Stands for Abstract Windowing Toolkit.
- Java AWT (Abstract Window Toolkit) is an API to develop Graphical User Interface (GUI) or windows-based applications in Java.
- AWT is used to develop Desktop Application.
- Java AWT components are platform-dependent i.e. components are displayed according to the view of operating system.
- AWT is heavy weight.
- The `java.awt` package provides classes and interfaces for AWT API.

GUI [Graphical User interface]



Java AWT Hierarchy





AWT Components

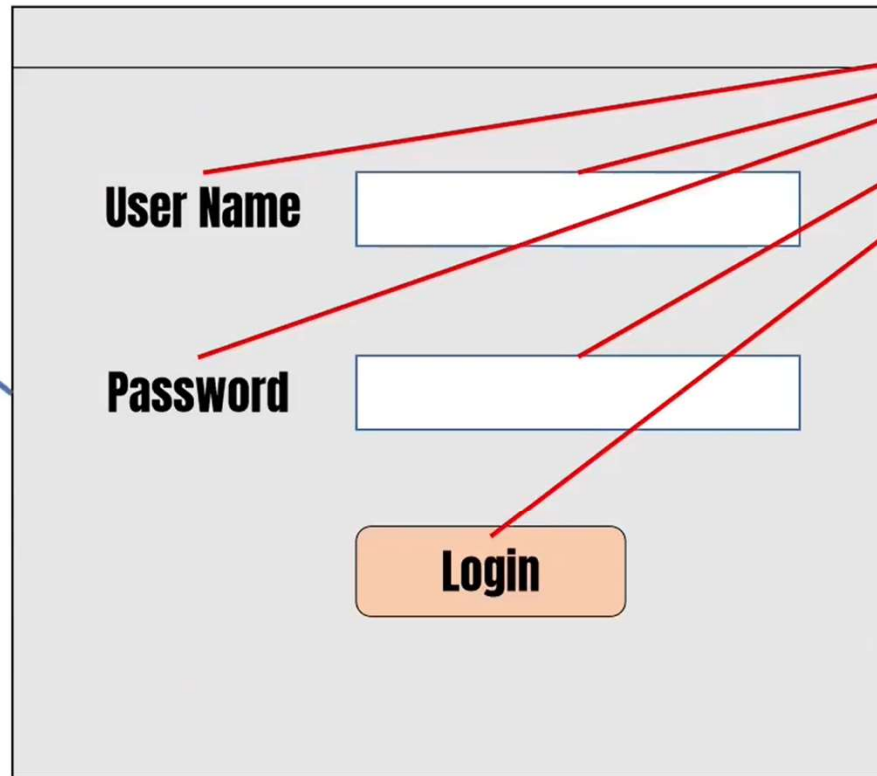
- To prepare GUI applications, AWT is able to provide support for the following components.

1. Label
2. Button
3. CheckBox
4. TextField
5. TextArea ... etc

- To represent the above GUI components, java.awt package has provided predefined classes.

GUI [Graphical User interface]

Container



Components

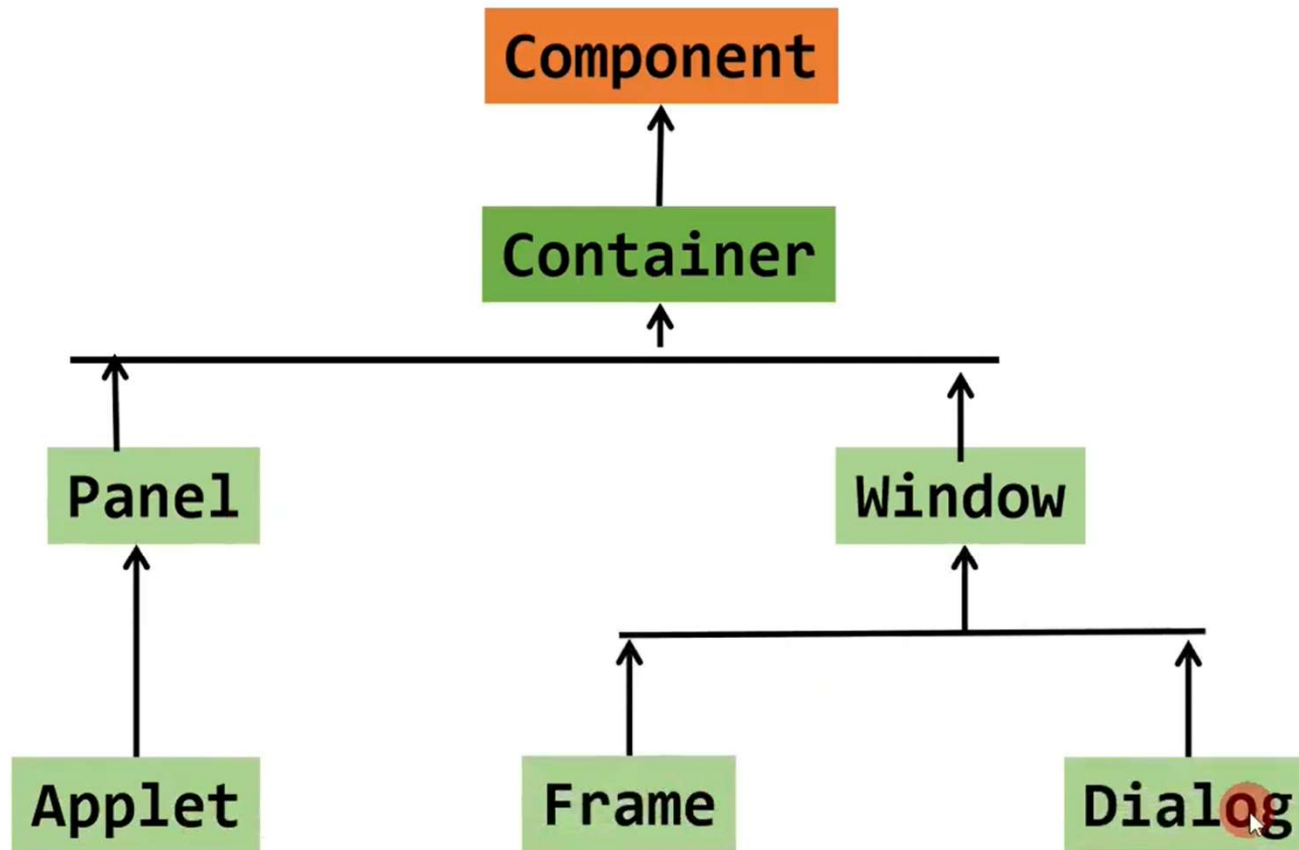
AWT Containers

- A container provides a space where a component can be located.

Definition

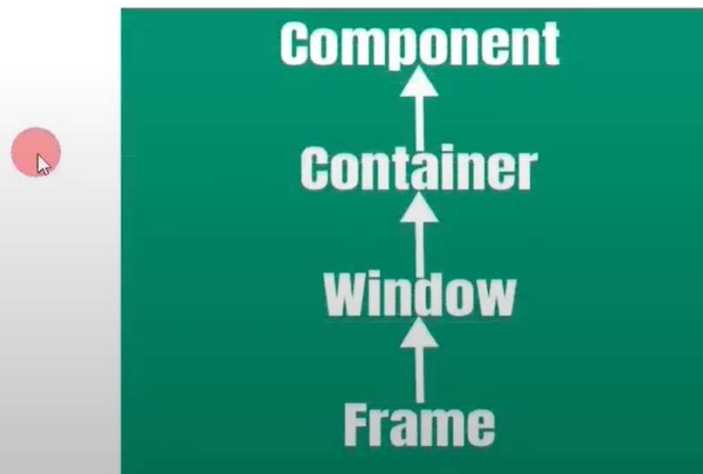
- A Container in AWT is a component itself and it adds the capability to add components[buttons, labels, textfield] to itself. Sub classes of Container are called as Containter. For example Panel, Frame and Window.

subclasses of container class in java



java.awt.Frame class

- Frame is a container, it able to manage some other GUI components like button, text field, scrollbar etc. .
- The Frame is the container that contain title bar and border and can have menu bars.
- To represent Frame in GUI applications, JAVA has provided a separate predefined class in the form java.awt.Frame.
- Frame is most widely used container while developing an AWT application.



Declaration

```
public class Frame extends Window implements MenuContainer  
{  
  
}
```

Constructors Frame class

```
public Frame():
```

It will create a Frame without title.

```
public Frame(String title):
```

It will create a Frame with the specified title.

Methods Frame class

```
public void setVisible(boolean b):
```

If b is true then it will provide visiblensess to the frame,
if b is false then it will hide the frame.

```
public void setSize(int width, int height):
```

In GUI applications, when we create Frame then Frame will
created with 0(zero) width and 0(zero) height, where to provide
size to the Frame explicitly we have to use setSize() method.

```
public void setTitle(String title):
```

To set a particular title to the Frame explicitly
we have to use setTitle() method.



There are two ways to create frame :-

- 1.By creating the object of Frame class (association)**
- 2.By extending Frame class (inheritance)**



2.By extending Frame class (inheritance)

In GUI applications, By creating the object of Frame class (association) is not suggestible to create Frames, because,we are unable to perform customization over the Frame. To implement customization over the Frame, we have to create user defined Frames.

Steps to create Frame by extending Frame class i.e user defined Frame

1. Declare an user defined class.
2. Extend User Defined Class from `java.awt.Frame` class.
3. Declare zero-argument constructor in user defined class.
4. Provide all Frame properties in User Defined Class constructor.
5. In main class,in `main()` method, create object for user defined Frame class.

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java.awt.Color class

- The Color class is a part of Java Abstract Window Toolkit(AWT) package.
- The Color class creates color by using the given RGBA values where RGBA stands for RED, GREEN, BLUE, ALPHA
- The value for individual components RGBA ranges from 0 to 255 or 0.0 to 1.0.
- The value of alpha determines the opacity of the color, where 0 or 0.0 stands fully transparent and 255 or 1.0 stands opaque.

Class Declaration

```
public class Color extends Object implements Paint, Serializable
{
    ---
    ---
}
```

Constants in Color class

```
public static final Color black: The color black
public static final Color BLACK: The color black
public static final Color blue: The color blue
public static final Color BLUE: The color blue
public static final Color red:The color red
public static final Color RED:The color red
public static final Color gray: The color gray
public static final Color GRAY: The color gray
public static final Color orange: The color orange
public static final Color ORANGE:The color orange
public static final Color yellow: The color yellow
public static final Color pink: The color pink
```


Constructors of Color class

`public Color(int red, int green, int blue):`

Creates RGB color with the specified red, green, and blue values in the range (0 - 255).

Example:-

`Color c=new Color(255,0,0) --->RED color`

`public Color(int red, int green, int blue, int alpha):`

Creates RGB color with the specified red, green, blue, and alpha values in the range (0 - 255).

Example:-

`Color c=new Color(255,0,0,255)`

```
public Color(float red, float green, float blue, float alpha):
```

Creates RGB color with the specified red, green, blue and alpha values in the range (0.0 - 1.0).

Example:-

```
Color c=new Color(1.0,0.0,0.0,1.0)
```

```
public Color(int rgb):
```

Creates RGB color with the specified combined RGB value

How to change the bgcolor of awt frame

To set background color of the Frame , we have to use the following method from java.awt.Frame class.

```
public void setBackground(Color clr)
```

Example1:-

```
Frame fm=new Frame();
```

```
Color c=Color.red;
```

```
fm.setBackground(c);
```

+ Example2:-

```
Frame fm=new Frame();
```

```
Color c=new Color(255,0,0)
```

```
fm.setBackground(c);
```

Commonly used methods of Graphics class

`public abstract void drawString(String str, int x, int y):`

It is used to draw the specified string / text.

Example: `g.drawString("Welcome to Uttamsoft", 50,50);`

`public void drawRect(int x, int y, int width, int height):`

It is used draws a rectangle with the specified width and height.

Example: `g.drawRect(50,50, 70,30);`

Commonly used methods of Graphics class

`public abstract void drawString(String str, int x, int y):`

It is used to draw the specified string / text.

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`public void drawRect(int x, int y, int width, int height):`

It is used draws a rectangle with the specified width and height

Example: `g.drawRect(50,50, 70,30);`

`if width equal to height`

It is used draws a Square with the specified width and height

Example: `g.drawRect(50,50, 70,70);`

`public abstract void fillRect(int x, int y, int width, int height):`

It is used to fill rectangle with the default color and specified width and height.

Commonly used methods of Graphics class

```
public abstract void drawArc(int x, int y, int width, int height,  
int startAngle, int arcAngle):
```

It is used draw a circular or elliptical arc.

Example: `g.drawArc(220, 120, 80, 40, 90, -90);`

```
public abstract void fillArc(int x, int y, int width, int height, int startAngle,  
int arcAngle):
```

It is used to fill a circular or elliptical arc.

Example: `g.fillArc(320, 120, 80, 40, 90, -90);`

```
public abstract void setColor(Color c):
```

It is used to set the graphics current color to the specified color.